

Rigorous Derivation of Emerging Dynamics in Background-Independent Qubit Networks

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1 Introduction to the Global Null Space

The fundamental premise of our model is the existence of a global state $|\Psi\rangle$ in the total Hilbert space $\mathcal{H}_{tot} = \mathcal{H}_C \otimes \mathcal{H}_S$. For this mapping to be mathematically valid, the total Hamiltonian $\hat{H}_{tot}$ must possess a non-trivial kernel (Null Space):

$$\text{Ker}(\hat{H}_{tot}) = \{|\Psi\rangle \in \mathcal{H}_{tot} \mid \hat{H}_{tot} |\Psi\rangle = 0\} \neq \emptyset. \quad (1)$$

Proposition 1.1. *Condition (1) holds if and only if the intersection of the energy spectra satisfies*

$$\text{Spec}(\hat{H}_C) \cap \text{Spec}(-\hat{H}_S) \neq \emptyset. \quad (2)$$

Proof. Let $\{E_k^C, |e_k\rangle_C\}$ and $\{E_j^S, |f_j\rangle_S\}$ be complete eigensystems of $\hat{H}_C$ and $\hat{H}_S$, respectively. Since $\hat{H}_{tot} = \hat{H}_C \otimes I_S + I_C \otimes \hat{H}_S$, the product basis $\{|e_k\rangle_C \otimes |f_j\rangle_S\}_{k,j}$ diagonalises $\hat{H}_{tot}$:

$$\hat{H}_{tot}(|e_k\rangle_C \otimes |f_j\rangle_S) = (E_k^C + E_j^S)(|e_k\rangle_C \otimes |f_j\rangle_S).$$

($\Rightarrow$) If $E_{k_0}^C + E_{j_0}^S = 0$, then $|e_{k_0}\rangle_C \otimes |f_{j_0}\rangle_S \neq 0$ is a non-trivial element of $\text{Ker}(\hat{H}_{tot})$.

($\Leftarrow$) Let $|\Psi\rangle = \sum_{k,j} c_{kj} |e_k\rangle_C \otimes |f_j\rangle_S \neq 0$ satisfy $\hat{H}_{tot} |\Psi\rangle = 0$. Then $\sum_{k,j} c_{kj} (E_k^C + E_j^S) |e_k\rangle_C \otimes |f_j\rangle_S = 0$. By linear independence of the product basis, $c_{kj} (E_k^C + E_j^S) = 0$ for all k, j . Since $|\Psi\rangle \neq 0$ there exists (k_0, j_0) with $c_{k_0 j_0} \neq 0$, forcing $E_{k_0}^C = -E_{j_0}^S \in \text{Spec}(\hat{H}_C) \cap \text{Spec}(-\hat{H}_S)$. $\square$

In an N -qubit simulation, condition (2) is achieved by tuning the local bias terms to ensure $E_k^C + E_j^S = 0$.

2 Clock Representation and Stone's Theorem

We define the *clock basis* $|t\rangle_C$ through the unitary group generated by $\hat{H}_C$. By Stone's theorem on one-parameter unitary groups, the strongly continuous map $t \mapsto e^{-i\hat{H}_C t}$ has $\hat{H}_C$ as its unique self-adjoint generator:

$$|t\rangle_C = U(t) |0\rangle_C = e^{-i\hat{H}_C t} |0\rangle_C. \quad (3)$$

Taking the Hilbert-space adjoint of (3) gives $\langle t|_C = \langle 0|_C e^{i\hat{H}_C t}$. Differentiating with respect to t :

$$\frac{d}{dt} \langle t|_C = \langle 0|_C \cdot i\hat{H}_C e^{i\hat{H}_C t} = \langle 0|_C e^{i\hat{H}_C t} \cdot i\hat{H}_C = \langle t|_C (i\hat{H}_C), \quad (4)$$

where the second equality uses $[e^{i\hat{H}_C t}, \hat{H}_C] = 0$; this commutator vanishes because $e^{i\hat{H}_C t} = \sum_{n=0}^{\infty} (i\hat{H}_C t)^n / n!$ and every power of $\hat{H}_C$ commutes with $\hat{H}_C$. Multiplying (4) by i :

$$i \frac{d}{dt} \langle t|_C = -\langle t|_C \hat{H}_C. \quad (5)$$

3 Step-by-Step Flow of Unitary Emergence

We now derive the internal Schrödinger equation from the static WdW equation.

Step 1: Projection of the Global Constraint

Starting from the WdW equation $\hat{H}_{tot} |\Psi\rangle = 0$, we project onto the clock state $\langle t|_C$:

$$\langle t|_C (\hat{H}_C \otimes I_S + I_C \otimes \hat{H}_S) |\Psi\rangle = 0. \quad (6)$$

Step 2: Separation of Operators

Expanding (6) by linearity:

$$\langle t|_C \hat{H}_C |\Psi\rangle + \langle t|_C (I_C \otimes \hat{H}_S) |\Psi\rangle = 0. \quad (7)$$

To show that $\hat{H}_S$ passes through the clock projection, write $|\Psi\rangle = \sum_\alpha |\phi_\alpha\rangle_C \otimes |\chi_\alpha\rangle_S$. Then

$$\langle t|_C (I_C \otimes \hat{H}_S) |\Psi\rangle = \sum_\alpha \langle t|\phi_\alpha|t|\phi_\alpha\rangle_C \hat{H}_S |\chi_\alpha\rangle_S = \hat{H}_S \left(\sum_\alpha \langle t|\phi_\alpha|t|\phi_\alpha\rangle_C |\chi_\alpha\rangle_S \right) = \hat{H}_S (\langle t|_C |\Psi\rangle),$$

where the second equality uses the $\mathbb{C}$ -linearity of $\hat{H}_S$ on $\mathcal{H}_S$ (the scalar coefficients $\langle t|\phi_\alpha|t|\phi_\alpha\rangle_C \in \mathbb{C}$ factor out). Equation (7) becomes:

$$\langle t|_C \hat{H}_C |\Psi\rangle + \hat{H}_S (\langle t|_C |\Psi\rangle) = 0. \quad (8)$$

Step 3: Identification of the Relative State

Define the *relative (conditional) state* $|\psi(t)\rangle_S \equiv \langle t|_C |\Psi\rangle$. Since $|\Psi\rangle$ is the time-independent WdW state, the time derivative passes through as:

$$i \frac{d}{dt} |\psi(t)\rangle_S = i \frac{d}{dt} (\langle t|_C |\Psi\rangle) = \left(i \frac{d}{dt} \langle t|_C \right) |\Psi\rangle \stackrel{(5)}{=} -\langle t|_C \hat{H}_C |\Psi\rangle. \quad (9)$$

Rearranging (9) gives $\langle t|_C \hat{H}_C |\Psi\rangle = -i \frac{d}{dt} |\psi(t)\rangle_S$. Substituting into (8):

$$-i \frac{d}{dt} |\psi(t)\rangle_S + \hat{H}_S |\psi(t)\rangle_S = 0. \quad (10)$$

Step 4: Final Derivation

Rearranging (10) yields the standard Schrödinger form:

$$i \frac{d}{dt} |\psi(t)\rangle_S = \hat{H}_S |\psi(t)\rangle_S. \quad (11)$$

This is the central result of the Page–Wootters mechanism: unitary time evolution emerges from the timeless WdW constraint $\hat{H}_{tot} |\Psi\rangle = 0$ via clock–system entanglement.

4 Mathematical Consistency Check: The Pauli Limit

In finite-dimensional qubit systems, the spectrum of $\hat{H}_C$ is discrete. Two important consequences follow.

Lemma 4.1 (Time Resolution). *For a clock Hamiltonian $\hat{H}_C$ with operator norm $\|\hat{H}_C\|$, the minimum resolvable time interval satisfies*

$$\Delta t \geq \frac{1}{2\|\hat{H}_C\|}, \quad \text{i.e. } \Delta t \sim 1/\|\hat{H}_C\|.$$

Proof. The Mandelstam–Tamm energy-time uncertainty relation states that for any observable $\hat{A}$:

$$\Delta E_C \cdot \tau_A \geq \frac{1}{2} |\langle [\hat{H}_C, \hat{A}] \rangle|,$$

where $\tau_A = \Delta A / |\langle \dot{\hat{A}} \rangle|$ is the characteristic evolution time of $\hat{A}$. Taking $\hat{A}$ to be the projector onto the initial clock state $|0\rangle_C$, τ_A equals the minimum time for the clock to become distinguishable from its initial value. Since $\Delta E_C \leq \|\hat{H}_C\|$, we obtain $\tau_A \geq (2\|\hat{H}_C\|)^{-1}$, i.e. $\Delta t \equiv \tau_A \sim 1/\|\hat{H}_C\|$. $\square$

Lemma 4.2 (Poincaré Recurrence). *The relative state $|\psi(t)\rangle_S$ satisfying (11) is quasi-periodic with recurrence time $T \leq 2\pi/\Delta E_{\min}$, where $\Delta E_{\min} := \min_{k \neq j} |E_k^S - E_j^S|$.*

Proof. Expand in the energy eigenbasis of $\hat{H}_S$:

$$|\psi(t)\rangle_S = \sum_j a_j e^{-iE_j^S t} |f_j\rangle_S, \quad a_j = \langle f_j | \psi(0) \rangle_S.$$

This is a finite sum of oscillatory terms (spectrum discrete by assumption). The state returns near its initial value when all relative phases satisfy $e^{-i(E_k^S - E_j^S)T} \approx 1$ simultaneously. The slowest cycle is governed by $\Delta E_{\min}$, requiring $\Delta E_{\min} \cdot T = 2\pi$, hence $T \leq 2\pi/\Delta E_{\min}$. $\square$

1. **Discretization:** By Lemma 4.1, time t is sampled at resolutions $\Delta t \sim 1/\|\hat{H}_C\|$.

2. **Poincaré Recurrence:** By Lemma 4.2, the state $|\psi(t)\rangle_S$ returns to its initial value after $T = 2\pi/\Delta E_{\min}$.

This ‘error’ is not a logical flaw but a physical consequence of finite information density (Bekenstein bound) in qubit-based universes.

5 Noise as Graph Perturbation

In the BI-QMN framework, the system is represented by a weighted graph $G = (V, E, W)$, where V is the set of qubits and W_{ij} represents the mutual information (or entanglement measure) between qubits i and j .

Definition 5.1 (Graph Weight Function). Given the N -qubit density matrix ρ , define the edge weight between qubits $i \neq j$ as $W_{ij} = |\rho_{ij}|$ (magnitude of off-diagonal coherence). The degree matrix is $D_{ii} = \sum_{j \neq i} W_{ij}$, and the Graph Laplacian is

$$L = D - W = \sum_{(i,j) \in E} W_{ij} (e_i - e_j)(e_i - e_j)^T, \quad (12)$$

where $e_k \in \mathbb{R}^{|V|}$ is the k -th standard basis vector.

Under a Lindblad decoherence channel, off-diagonal elements decay as $\rho_{ij}(t) = \rho_{ij}(0)e^{-\gamma_{ij}t}$, so for a time step Δt :

$$\delta W_{ij} = W_{ij}(0)(1 - e^{-\gamma_{ij}\Delta t}) \approx W_{ij}(0)\gamma_{ij}\Delta t > 0.$$

Each weight reduction contributes the rank-one perturbation $-\delta W_{ij}(e_i - e_j)(e_i - e_j)^T$ to L , giving

$$L_{\text{noise}} = L_0 + \Delta L, \quad \Delta L = - \sum_{(i,j) \in E} \delta W_{ij} (e_i - e_j)(e_i - e_j)^T. \quad (13)$$

Since each rank-one summand is positive semidefinite and $\delta W_{ij} > 0$, ΔL is *negative semidefinite* ($\Delta L \preceq 0$).

6 Weyl's Inequality and Spectral Shift

Theorem 6.1 (Weyl's Inequality). *Let A and B be $n \times n$ Hermitian matrices with ordered eigenvalues. Then for all $i = 1, \dots, n$:*

$$\lambda_i(A) + \lambda_{\min}(B) \leq \lambda_i(A + B) \leq \lambda_i(A) + \lambda_{\max}(B).$$

Applying Theorem 6.1 with $A = L_0$ and $B = \Delta L$, the perturbed eigenvalues $\tilde{\lambda}_i$ satisfy:

$$\lambda_i(L_0) + \lambda_{\min}(\Delta L) \leq \tilde{\lambda}_i \leq \lambda_i(L_0) + \lambda_{\max}(\Delta L). \quad (14)$$

Because $\Delta L \preceq 0$, we have $\lambda_{\max}(\Delta L) \leq 0$, so the upper bound gives $\tilde{\lambda}_i \leq \lambda_i(L_0)$: *all eigenvalues shift downward*. Furthermore $\text{tr}(\Delta L) = -\sum_{i,j} \delta W_{ij} < 0$ for any non-zero noise, which forces $\lambda_{\min}(\Delta L) < 0$. The magnitude of the shift is bounded below by

$$|\lambda_{\min}(\Delta L)| = \|\Delta L\|_2 \geq \frac{1}{\sqrt{n}} \|\Delta L\|_F > 0,$$

where we used $\|B\|_2 \geq \|B\|_F / \sqrt{n}$ for Hermitian B . This confirms that any non-zero noise ΔL induces a deterministic, detectable shift in the spectral density, providing a mathematical basis for anomaly detection.

7 Error Differentiation: X vs. Z Errors

We hypothesize that different quantum error channels map to distinct types of structural distortions.

7.1 Bit-Flip (X) Errors as Edge Deletion

An X -error on qubit k reduces the effective entanglement with its neighbors, decreasing weights $\{W_{kj}\}_{j \sim k}$ by $\{\delta W_{kj}\}$.

Proposition 7.1 (First-Order Spectral Shift under X -Errors). *Under an X -error that reduces edge weights $\{W_{kj}\}$ by $\{\delta W_{kj} > 0\}$, the first-order shift of the m -th Laplacian eigenvalue is*

$$\delta \lambda_m = - \sum_{j \sim k} \delta W_{kj} ((v_m)_k - (v_m)_j)^2 \leq 0, \quad (15)$$

where $\mathbf{v}_m$ is the m -th normalised eigenvector of L_0 (assuming non-degeneracy).

Proof. From Eq. (12), edge (k, j) contributes $W_{kj}(e_k - e_j)(e_k - e_j)^T$ to L . A weight reduction by δW_{kj} induces the elementary perturbation $\Delta L_{(k,j)} = -\delta W_{kj}(e_k - e_j)(e_k - e_j)^T$. By first-order Rayleigh–Schrödinger perturbation theory for non-degenerate eigenvalues:

$$\begin{aligned} \delta \lambda_m &= \mathbf{v}_m^T \Delta L \mathbf{v}_m = - \sum_{j \sim k} \delta W_{kj} \mathbf{v}_m^T (e_k - e_j)(e_k - e_j)^T \mathbf{v}_m \\ &= - \sum_{j \sim k} \delta W_{kj} [(v_m)_k - (v_m)_j]^2. \end{aligned}$$

Non-negativity of the squared terms and positivity of δW_{kj} ensure $\delta \lambda_m \leq 0$. $\square$

For high-frequency eigenvectors $\mathbf{v}_m$ (large m), rapid spatial oscillation ensures $|(v_m)_k - (v_m)_j| = \mathcal{O}(1)$, so the aggregate shift across all error edges is:

$$\Delta \lambda_{\text{large}} \propto - \sum_{(i,j) \in \text{error}} \delta W_{ij}. \quad (16)$$

7.2 Phase-Flip (Z) Errors as Topological Warp

Z -errors disrupt the phase relationship $\langle t|_C|\Psi\rangle$, causing global connectivity loss. This is captured by the *Fiedler value* λ_2 , which bounds the edge expansion of G via the Cheeger inequality:

$$\frac{\lambda_2}{2} \leq h(G) \leq \sqrt{2\lambda_2 \lambda_{\max}(L)},$$

where $h(G)$ is the Cheeger constant. The perturbed Fiedler value is given by the Courant–Fischer min-max theorem:

$$\tilde{\lambda}_2 = \min_{\substack{x \perp \mathbf{1} \\ \|x\|=1}} \frac{x^T (L_0 + \Delta L) x}{x^T x}, \quad (17)$$

where $\mathbf{1} \in \mathbb{R}^n$ denotes the all-ones vector. Since $\Delta L \preceq 0$, we have $x^T \Delta L x \leq 0$ for all x , therefore

$$\tilde{\lambda}_2 \leq \min_{\substack{x \perp \mathbf{1} \\ \|x\|=1}} x^T L_0 x = \lambda_2.$$

The gap $\lambda_2 - \tilde{\lambda}_2 \geq 0$ directly quantifies the loss of algebraic connectivity (global phase coherence). A significant drop indicates *Phase Collapse*.

8 Relational Spectral Detection Without External Time

8.1 Clock-Conditioned Relative States

We do not assume an external background time parameter. Instead, following the Page–Wootters construction, we consider a timeless global state $|\Psi\rangle \in \mathcal{H}_C \otimes \mathcal{H}_S$ satisfying the global constraint

$$\hat{H}_{\text{tot}}|\Psi\rangle = 0. \quad (18)$$

Let $\{|\tau\rangle_C\}$ denote the clock basis. The corresponding clock-conditioned relative state is defined as

$$|\psi(\tau)\rangle_S := {}_C\langle\tau|\Psi\rangle, \quad (19)$$

or, in density-matrix form,

$$\rho_S(\tau) := \frac{\text{Tr}_C[(|\tau\rangle\langle\tau|_C \otimes I_S) |\Psi\rangle\langle\Psi|]}{\text{Tr}[(|\tau\rangle\langle\tau|_C \otimes I_S) |\Psi\rangle\langle\Psi|]}. \quad (20)$$

Here, τ is not an external time variable, but an internal relational label induced by the clock subsystem.

8.2 Relational Graph and Laplacian Family

For each clock-conditioned state $\rho_S(\tau)$, we construct a weighted graph

$$G_\tau = (V, E, W^{(\tau)}), \quad (21)$$

where the edge weights are determined from the relational structure of $\rho_S(\tau)$.

For example, one may define

$$W_{ij}^{(\tau)} := f(\rho_S(\tau); i, j), \quad (22)$$

where f denotes a pairwise relational measure such as coherence magnitude, mutual information, or another correlation functional.

The corresponding degree matrix is

$$D_{ii}^{(\tau)} = \sum_{j \neq i} W_{ij}^{(\tau)}, \quad (23)$$

and the clock-conditioned graph Laplacian is

$$L(\tau) = D^{(\tau)} - W^{(\tau)}. \quad (24)$$

Thus, instead of a time-evolving Laplacian, we obtain a relationally indexed family of Laplacians

$$\{L(\tau)\}_{\tau \in \mathcal{T}}, \quad (25)$$

where $\mathcal{T}$ denotes the clock-label domain.

8.3 Relational Spectral Density

Let

$$\Lambda(\tau) = \{\lambda_1(\tau), \dots, \lambda_n(\tau)\} \quad (26)$$

be the eigenvalue set of $L(\tau)$.

We define the clock-conditioned empirical spectral density (ESD) as

$$p_\tau(\lambda) = \frac{1}{n} \sum_{k=1}^n \delta(\lambda - \lambda_k(\tau)), \quad (27)$$

or, numerically, via a kernel density estimator,

$$p_\tau(\lambda) = \frac{1}{n} \sum_{k=1}^n \frac{1}{\sqrt{2\pi\sigma_h^2}} \exp\left(-\frac{(\lambda - \lambda_k(\tau))^2}{2\sigma_h^2}\right). \quad (28)$$

Accordingly, spectral variation is interpreted as variation across clock-conditioned relational slices, rather than evolution over an external time axis.

8.4 Reference and Observed Relational Spectra

For each relational slice τ , let $p_\tau^{\text{ref}}(\lambda)$ denote the reference spectral density and $p_\tau^{\text{obs}}(\lambda)$ the observed spectral density under perturbation.

The slice-wise information-theoretic discrepancy is defined as

$$D_{\text{KL}}^{(\tau)} := D_{\text{KL}}\left(p_\tau^{\text{ref}} \parallel p_\tau^{\text{obs}}\right). \quad (29)$$

Theorem 8.1 (Relational Spectral Detectability). *For a fixed clock-conditioned slice τ , if the perturbation induces a non-trivial spectral deformation, $\|\Delta L(\tau)\|_F > 0$, then $D_{\text{KL}}^{(\tau)} > 0$.*

Proof. If $\|\Delta L(\tau)\|_F > 0$, then by Weyl's inequality (Theorem 6.1) at least one eigenvalue of $L(\tau)$ is shifted relative to the reference Laplacian. Hence $p_\tau^{\text{ref}}(\lambda) \neq p_\tau^{\text{obs}}(\lambda)$ on a set of non-zero measure. By Gibbs' inequality, $D_{\text{KL}}(p_\tau^{\text{ref}} \parallel p_\tau^{\text{obs}}) > 0$. $\square$

Thus, $D_{\text{KL}}^{(\tau)}$ serves as a sufficient certificate of non-trivial spectral deformation within the relational slice indexed by τ .

8.5 Relationally Averaged Detection Functional

To compare the entire relational family rather than a single slice, we define the aggregated detection functional

$$\mathcal{D}_{\text{KL}} = \int_{\mathcal{T}} D_{\text{KL}}\left(p_\tau^{\text{ref}} \parallel p_\tau^{\text{obs}}\right) d\mu(\tau), \quad (30)$$

where $\mu(\tau)$ is a measure over the clock-label domain $\mathcal{T}$.

In discrete simulations, this reduces to

$$\mathcal{D}_{\text{KL}} = \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} D_{\text{KL}}^{(\tau)}. \quad (31)$$

This quantity measures the global mismatch between two relational spectral families without introducing an external time parameter.

8.6 Hybrid Detection Metric with Geometric Augmentation

While $D_{\text{KL}}^{(\tau)}$ provides an information-theoretic detection certificate, it does not directly quantify the geometric magnitude of spectral displacement. To incorporate geometric sensitivity, we define

$$D_{\text{detect}}^{(\tau)} = \alpha \tilde{D}_{\text{KL}}^{(\tau)} + (1 - \alpha) \widetilde{W}_1^{(\tau)}, \quad (32)$$

where

$$\tilde{D}_{\text{KL}}^{(\tau)} = \text{Norm} \left[D_{\text{KL}} \left(p_{\tau}^{\text{ref}} \parallel p_{\tau}^{\text{obs}} \right) \right] \quad (33)$$

and

$$\widetilde{W}_1^{(\tau)} = \text{Norm} \left[W_1 \left(p_{\tau}^{\text{ref}}, p_{\tau}^{\text{obs}} \right) \right]. \quad (34)$$

Here, W_1 denotes the Wasserstein-1 distance between the two clock-conditioned spectral densities, and $\alpha \in [0, 1]$ controls the balance between information-theoretic sensitivity and geometric displacement.

We then define the relationally aggregated hybrid score

$$\mathcal{D}_{\text{detect}} = \int_{\mathcal{T}} D_{\text{detect}}^{(\tau)} d\mu(\tau), \quad (35)$$

or, in discrete form,

$$\mathcal{D}_{\text{detect}} = \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} D_{\text{detect}}^{(\tau)}. \quad (36)$$

Remark 8.2. The KL term preserves the information-theoretic detectability guarantee at each relational slice, while the Wasserstein term quantifies geometric spectral displacement across the clock-conditioned family.

8.7 Symmetric Stability Check

For numerical robustness, we optionally introduce the Jensen–Shannon divergence at each relational slice:

$$D_{\text{JS}}^{(\tau)} = \frac{1}{2} D_{\text{KL}} \left(p_{\tau}^{\text{ref}} \parallel m_{\tau} \right) + \frac{1}{2} D_{\text{KL}} \left(p_{\tau}^{\text{obs}} \parallel m_{\tau} \right), \quad (37)$$

with

$$m_{\tau} = \frac{1}{2} \left(p_{\tau}^{\text{ref}} + p_{\tau}^{\text{obs}} \right). \quad (38)$$

The relationally averaged Jensen–Shannon score is

$$\mathcal{D}_{\text{JS}} = \int_{\mathcal{T}} D_{\text{JS}}^{(\tau)} d\mu(\tau), \quad (39)$$

or its discrete counterpart. This quantity is used as a symmetric numerical stability check, rather than as the primary detectability certificate.

8.8 Localized Geometric Error Differentiation

To distinguish different structural perturbation types, we introduce a band-limited Wasserstein descriptor on a spectral region $\mathcal{B}$:

$$W_1^{(\tau, \mathcal{B})} = \int_{\mathcal{B}} \left| F_{\tau}^{\text{ref}}(\lambda) - F_{\tau}^{\text{obs}}(\lambda) \right| d\lambda, \quad (40)$$

where F_{τ}^{ref} and F_{τ}^{obs} are the corresponding cumulative spectral distributions.

This enables local-versus-global deformation analysis within each relational slice:

- high-frequency bands capture localized structural perturbations,
- low-frequency bands capture global connectivity loss.

Thus, error discrimination is formulated as a geometric comparison across relational spectral slices rather than as a trajectory over background time.

8.9 Detection Criterion

An anomaly is declared whenever $\mathcal{D}_{\text{detect}} > \varepsilon$, where ε is determined from the background distribution of relationally aggregated divergence values under unperturbed fluctuations:

$$\varepsilon = \mathbb{E}[\mathcal{D}_{\text{vac}}] + \kappa \text{Std}[\mathcal{D}_{\text{vac}}]. \quad (41)$$

This criterion remains fully relational and does not require any external time indexing.

9 Relational Spectral Trajectory Manifold

9.1 Clock-Conditioned Spectral Family

Let $|\Psi\rangle \in \mathcal{H}_C \otimes \mathcal{H}_S$ be a timeless global state satisfying

$$\hat{H}_{\text{tot}}|\Psi\rangle = 0. \quad (42)$$

For each clock label $\tau \in \mathcal{T}$, define the clock-conditioned relative state

$$\rho_S(\tau) := \frac{\text{Tr}_C[(|\tau\rangle\langle\tau|_C \otimes I_S) |\Psi\rangle\langle\Psi|]}{\text{Tr}[(|\tau\rangle\langle\tau|_C \otimes I_S) |\Psi\rangle\langle\Psi|]}. \quad (43)$$

From each relational slice $\rho_S(\tau)$, construct the weighted graph Laplacian

$$L(\tau) = D^{(\tau)} - W^{(\tau)}, \quad (44)$$

and let its ordered eigenvalue vector be

$$\Lambda(\tau) = (\lambda_1(\tau), \lambda_2(\tau), \dots, \lambda_n(\tau)) \in \mathbb{R}^n. \quad (45)$$

We define the *relational spectral family* associated with $|\Psi\rangle$ as

$$\Gamma_\Psi : \mathcal{T} \rightarrow \mathbb{R}^n, \quad \Gamma_\Psi(\tau) := \Lambda(\tau). \quad (46)$$

Thus, Γ_Ψ is not a trajectory over external time, but an ordered family of spectral signatures indexed by the internal clock labels.

9.2 Spectral Trajectory Space

Let $\mathcal{M}_{\text{spec}}$ denote the set of admissible relational spectral families,

$$\mathcal{M}_{\text{spec}} := \{\Gamma : \mathcal{T} \rightarrow \mathbb{R}^n \mid \Gamma(\tau) = \Lambda(L(\rho_S(\tau))) \text{ for some physically admissible } \rho_S(\tau)\}. \quad (47)$$

We interpret $\mathcal{M}_{\text{spec}}$ as the *relational spectral trajectory manifold*, i.e., the manifold of all physically realizable clock-conditioned spectral families.

Remark 9.1. Each element of $\mathcal{M}_{\text{spec}}$ is a full relational trajectory Γ , not a single spectral snapshot.

Remark 9.2. Strictly speaking, $\mathcal{M}_{\text{spec}}$ is treated here as a metric trajectory space of admissible spectral families rather than as a fully established smooth manifold. The geodesic formulation in Secs. 9.3–9.9 should therefore be interpreted as a formal geometric model. The discrete representation in Eq. (84) provides the computational realization, in which $\mathcal{M}_{\text{spec}}$ reduces to a constrained subset of $\mathbb{R}^{mn}$.

9.3 Tangent Structure

Given a smooth one-parameter family of relational trajectories

$$\Gamma_s \in \mathcal{M}_{\text{spec}}, \quad s \in (-\epsilon, \epsilon), \quad (48)$$

with $\Gamma_0 = \Gamma$, the tangent vector at Γ is defined by

$$V(\tau) = \left. \frac{d}{ds} \Gamma_s(\tau) \right|_{s=0} \in \mathbb{R}^n. \quad (49)$$

Hence, the tangent space at Γ is

$$T_\Gamma \mathcal{M}_{\text{spec}} = \left\{ V : \mathcal{T} \rightarrow \mathbb{R}^n \mid V(\tau) = \left. \frac{d}{ds} \Gamma_s(\tau) \right|_{s=0} \text{ for some smooth family } \Gamma_s \right\}. \quad (50)$$

Intuitively, V describes an infinitesimal deformation of the full relational spectral family.

9.4 Riemannian Metric on the Trajectory Manifold

To quantify distances between relational trajectories, we introduce a Riemannian metric on $\mathcal{M}_{\text{spec}}$.

For tangent vectors $U, V \in T_\Gamma \mathcal{M}_{\text{spec}}$, define

$$g_\Gamma(U, V) = \int_{\mathcal{T}} U(\tau)^\top G(\Gamma(\tau)) V(\tau) d\mu(\tau), \quad (51)$$

where $G(\Gamma(\tau))$ is a positive-definite local metric tensor on $\mathbb{R}^n$, and μ is a measure over the clock-label domain $\mathcal{T}$.

A simple Euclidean choice is

$$G(\Gamma(\tau)) = I_n, \quad (52)$$

which yields

$$g_\Gamma(U, V) = \int_{\mathcal{T}} U(\tau)^\top V(\tau) d\mu(\tau). \quad (53)$$

More generally, one may choose G to weight low-frequency and high-frequency spectral components differently.

9.5 Trajectory Length and Distance

Let $\gamma : [0, 1] \rightarrow \mathcal{M}_{\text{spec}}$ be a smooth path in the trajectory manifold, where each $\gamma(s)$ is itself a relational spectral family:

$$\gamma(s) = \Gamma_s. \quad (54)$$

Its velocity field is

$$\dot{\gamma}(s) = \frac{d}{ds} \Gamma_s \in T_{\Gamma_s} \mathcal{M}_{\text{spec}}. \quad (55)$$

The length of γ is defined by

$$\text{Len}(\gamma) = \int_0^1 \sqrt{g_{\Gamma_s}(\dot{\gamma}(s), \dot{\gamma}(s))} ds. \quad (56)$$

The induced geodesic distance between two relational trajectories $\Gamma^{(a)}, \Gamma^{(b)} \in \mathcal{M}_{\text{spec}}$ is

$$d_{\mathcal{M}}(\Gamma^{(a)}, \Gamma^{(b)}) = \inf_{\gamma(0)=\Gamma^{(a)}, \gamma(1)=\Gamma^{(b)}} \text{Len}(\gamma). \quad (57)$$

9.6 Observed and Reference Trajectories

Let

$$\Gamma_{\text{ref}} \in \mathcal{M}_{\text{spec}} \quad (58)$$

denote the reference relational spectral trajectory, and let

$$\Gamma_{\text{obs}} \in \mathcal{M}_{\text{spec}} \quad (59)$$

denote the observed trajectory under perturbation.

The trajectory-level distortion is then defined as

$$\Delta_{\text{traj}} = d_{\mathcal{M}}(\Gamma_{\text{ref}}, \Gamma_{\text{obs}}). \quad (60)$$

This quantity measures the global deformation of the entire clock-conditioned spectral family, rather than the discrepancy at any single relational slice.

9.7 Geodesic Recovery Principle

We now formulate recovery as a geometric restoration problem on $\mathcal{M}_{\text{spec}}$.

Given an observed trajectory Γ_{obs} , define the recovered trajectory as

$$\Gamma^* = \arg \min_{\Gamma \in \mathcal{A}} d_{\mathcal{M}}(\Gamma, \Gamma_{\text{obs}}), \quad (61)$$

where $\mathcal{A} \subset \mathcal{M}_{\text{spec}}$ is the admissible set of physically consistent or reference-compatible trajectories.

Equivalently, if Γ_{ref} is known, the recovery path is the minimizing geodesic

$$\gamma^* = \arg \min_{\gamma(0)=\Gamma_{\text{obs}}, \gamma(1)=\Gamma_{\text{ref}}} \text{Len}(\gamma). \quad (62)$$

This gives the central interpretation:

Recovery is not modeled as evolution over background time, but as the shortest admissible path on the relational spectral trajectory manifold.

9.8 Geodesic Equation

Let the path be locally represented by coordinates

$$x^a(s), \quad a = 1, \dots, \dim(\mathcal{M}_{\text{spec}}), \quad (63)$$

with Riemannian metric $g_{ab}(x)$. The geodesic equation is

$$\frac{d^2 x^a}{ds^2} + \Gamma_{bc}^a(x) \frac{dx^b}{ds} \frac{dx^c}{ds} = 0, \quad (64)$$

where the Christoffel symbols are

$$\Gamma_{bc}^a = \frac{1}{2} g^{ad} (\partial_b g_{dc} + \partial_c g_{db} - \partial_d g_{bc}). \quad (65)$$

In the Euclidean metric case $G = I_n$, the connection is flat and the geodesic reduces to a straight interpolation in trajectory space:

$$\Gamma_s(\tau) = (1-s)\Gamma_{\text{obs}}(\tau) + s\Gamma_{\text{ref}}(\tau), \quad s \in [0, 1]. \quad (66)$$

9.9 Variational Formulation

Equivalently, geodesics are stationary points of the action

$$\mathcal{S}[\gamma] = \frac{1}{2} \int_0^1 g_{\Gamma_s}(\dot{\gamma}(s), \dot{\gamma}(s)) ds. \quad (67)$$

Thus, the geodesic recovery problem can be formulated as

$$\gamma^* = \arg \min_{\gamma} \mathcal{S}[\gamma], \quad (68)$$

subject to the endpoint conditions

$$\gamma(0) = \Gamma_{\text{obs}}, \quad \gamma(1) = \Gamma_{\text{ref}}, \quad (69)$$

or more generally,

$$\gamma(1) \in \mathcal{A}. \quad (70)$$

10 Admissible Relational Spectral Trajectory Set

10.1 Definition of the Admissible Set

Let $\mathcal{M}_{\text{spec}}$ denote the relational spectral trajectory manifold. We define the admissible set $\mathcal{A} \subset \mathcal{M}_{\text{spec}}$ as

$$\mathcal{A} = \mathcal{A}_{\text{lap}} \cap \mathcal{A}_{\text{rel}} \cap \mathcal{A}_{\text{ref}}, \quad (71)$$

where each component enforces a distinct class of physical and structural constraints.

10.2 Laplacian Consistency

A trajectory $\Gamma \in \mathcal{M}_{\text{spec}}$ belongs to $\mathcal{A}_{\text{lap}}$ if for all $\tau \in \mathcal{T}$, the associated matrix $L(\tau)$ satisfies

$$L(\tau) = L(\tau)^\top, \quad (72)$$

$$L(\tau) \succeq 0, \quad (73)$$

$$L(\tau)\mathbf{1} = 0, \quad (74)$$

$$L_{ij}(\tau) \leq 0 \quad (i \neq j), \quad (75)$$

$$L_{ii}(\tau) = \sum_{j \neq i} -L_{ij}(\tau). \quad (76)$$

Thus,

$$\mathcal{A}_{\text{lap}} = \{\Gamma \in \mathcal{M}_{\text{spec}} \mid L(\tau) \text{ is a valid graph Laplacian for all } \tau\}. \quad (77)$$

10.3 Relational Consistency

A trajectory Γ is relationally admissible if it satisfies

(i) Slice Regularity

$$\int_{\mathcal{T}} \left\| \frac{d}{d\tau} \Gamma(\tau) \right\|^2 d\mu(\tau) < \infty. \quad (78)$$

(ii) Spectral Ordering

$$\lambda_1(\tau) \leq \lambda_2(\tau) \leq \dots \leq \lambda_n(\tau) \quad \forall \tau \in \mathcal{T}. \quad (79)$$

(iii) **Clock Consistency** Let $d_{\text{clock}}(\rho_S(\tau))$ be a clock-consistency measure. Then

$$\int_{\mathcal{T}} d_{\text{clock}}(\rho_S(\tau)) d\mu(\tau) \leq \eta. \quad (80)$$

Thus,

$$\mathcal{A}_{\text{rel}} = \{\Gamma \in \mathcal{A}_{\text{lap}} \mid \Gamma \text{ is slice-regular and clock-consistent}\}. \quad (81)$$

10.4 Reference Compatibility

Let $\mathfrak{R} \subset \mathcal{M}_{\text{spec}}$ be a reference trajectory family. We define

$$\mathcal{A}_{\text{ref}} = \left\{ \Gamma \in \mathcal{A}_{\text{rel}} \mid \inf_{\Gamma' \in \mathfrak{R}} d_{\mathcal{M}}(\Gamma, \Gamma') \leq \delta \right\}. \quad (82)$$

10.5 Final Definition

The admissible set is given by

$$\mathcal{A} = \left\{ \Gamma \in \mathcal{M}_{\text{spec}} \mid \begin{array}{l} L(\tau) \text{ is a valid Laplacian for all } \tau, \\ \Gamma \text{ is slice-regular and clock-consistent,} \\ \inf_{\Gamma' \in \mathfrak{R}} d_{\mathcal{M}}(\Gamma, \Gamma') \leq \delta \end{array} \right\}. \quad (83)$$

10.6 Discrete Computational Form

Let $\mathcal{T} = \{\tau_1, \dots, \tau_m\}$. Then a trajectory is represented as

$$\mathbf{z}_{\Gamma} = \begin{bmatrix} \Lambda(\tau_1) \\ \Lambda(\tau_2) \\ \vdots \\ \Lambda(\tau_m) \end{bmatrix} \in \mathbb{R}^{mn}. \quad (84)$$

The trajectory distance becomes

$$d_{\mathcal{M}}(\Gamma^{(a)}, \Gamma^{(b)}) = \left(\sum_{r=1}^m w_r \left\| \Lambda^{(a)}(\tau_r) - \Lambda^{(b)}(\tau_r) \right\|_2^2 \right)^{1/2}. \quad (85)$$

10.7 Penalty-Based Admissibility

Define

$$\Phi_{\text{lap}}(\Gamma) = \sum_{r=1}^m \left(\|L(\tau_r) - L(\tau_r)^{\top}\|_F^2 + \|\min(0, \lambda(L(\tau_r)))\|_2^2 + \|L(\tau_r)\mathbf{1}\|_2^2 \right), \quad (86)$$

$$\Phi_{\text{smooth}}(\Gamma) = \sum_{r=1}^{m-1} \frac{\|\Lambda(\tau_{r+1}) - \Lambda(\tau_r)\|_2^2}{(\tau_{r+1} - \tau_r)^2}, \quad (87)$$

$$\Phi_{\text{ref}}(\Gamma) = \min_{\Gamma' \in \mathfrak{R}} d_{\mathcal{M}}(\Gamma, \Gamma')^2, \quad (88)$$

$$\Phi_{\text{clock}}^{\text{geom}}(\Gamma) = \sum_{r=1}^{m-1} \frac{\|\Lambda(\tau_{r+1}) - \Lambda(\tau_r)\|_2}{d_{\text{clock}}^{\text{geom}}(\tau_r, \tau_{r+1}) + \epsilon} \quad (89)$$

Here Φ_{lap} is a graph-validity surrogate, Φ_{smooth} is a regularizer on relational variation, and Φ_{clock} is a relational consistency penalty. The quantity Φ_{ref} remains useful as a coarse reference-compatibility diagnostic, but it does not by itself provide a directed restoration force toward the clean family.

We therefore define the relaxed admissible set

$$\mathcal{A}_\epsilon = \{\Gamma : \Phi_{\text{lap}} \leq \epsilon_1, \Phi_{\text{smooth}} \leq \epsilon_2, \Phi_{\text{ref}} \leq \epsilon_3, \Phi_{\text{clock}} \leq \epsilon_4\}. \quad (90)$$

10.8 Recovery Objective

The earlier recovery objective primarily enforced admissibility and observational consistency, but did not sufficiently encode restoration toward the reference family. To address this mismatch, we introduce an explicit reference-guided recovery term based on the nearest admissible reference anchor. This modification separates restoration from feasibility, allowing the optimization to favor trajectories that are both admissible and structurally closer to the clean reference family.

Let

$$\Gamma_{\text{ref,anchor}} := \arg \min_{\Gamma' \in \mathcal{R}} d_{\mathcal{M}}(\Gamma', \Gamma_{\text{obs}})^2 \quad (91)$$

denote the nearest reference anchor to the observed trajectory.

The recovered trajectory is then defined by the restoration-aware objective

$$\Gamma^* = \arg \min_{\Gamma} \left[\alpha d_{\mathcal{M}}(\Gamma, \Gamma_{\text{obs}})^2 + \beta d_{\mathcal{M}}(\Gamma, \Gamma_{\text{ref,anchor}})^2 + \lambda_1 \Phi_{\text{lap}}(\Gamma) + \lambda_2 \Phi_{\text{smooth}}(\Gamma) + \lambda_3 \Phi_{\text{clock}}(\Gamma) \right]. \quad (92)$$

In this formulation, the observed-fit term preserves consistency with the measured trajectory, the anchor term supplies an explicit restoration direction toward the reference family, and the remaining penalties serve only as feasibility / regularization controls.

10.9 Two-Stage Admissible Recovery

We refine the recovery procedure by introducing a two-stage structure on the admissible trajectory space.

Stage 1 (Discrete Admissible Projection). Given an observed trajectory Γ_{obs} , we first project onto a discrete admissible candidate bank $\mathcal{A}_{\text{bank}} \subset \mathcal{A}$:

$$\Gamma^{(1)} = \arg \min_{\Gamma \in \mathcal{A}_{\text{bank}}} \left[\alpha d_M(\Gamma, \Gamma_{\text{obs}})^2 + \beta d_M(\Gamma, \Gamma_{\text{ref,anchor}})^2 + \lambda_1 \Phi_{\text{lap}}(\Gamma) + \lambda_2 \Phi_{\text{smooth}}(\Gamma) + \lambda_3 \Phi_{\text{clock}}(\Gamma) \right].$$

This step provides a stable admissible anchor while explicitly preserving restoration toward the nearest compatible reference family.

Stage 2 (Admissible Convex Refinement). We optionally perform a continuous refinement within a local convex hull of admissible candidates. Let $\{\Gamma_k\}$ denote a selected subset of admissible candidates. We consider convex combinations

$$\Gamma(\alpha) = \sum_k \alpha_k \Gamma_k, \quad \alpha_k \geq 0, \quad \sum_k \alpha_k = 1,$$

restricted to those α such that $\Gamma(\alpha) \in \mathcal{A}$.

The refined trajectory is defined as

$$\Gamma^{(2)} = \arg \min_{\Gamma(\alpha) \in \mathcal{A}} \left[\alpha d_M(\Gamma(\alpha), \Gamma_{\text{obs}})^2 + \beta d_M(\Gamma(\alpha), \Gamma_{\text{ref,anchor}})^2 + \lambda_1 \Phi_{\text{lap}}(\Gamma(\alpha)) + \lambda_2 \Phi_{\text{smooth}}(\Gamma(\alpha)) + \lambda_3 \Phi_{\text{clock}}(\Gamma(\alpha)) \right].$$

Interpretation. Stage 1 performs a coarse projection onto an admissible discrete set, while Stage 2 carries out a local refinement under the same restoration-aware objective. The reference-anchor term is the actual recovery driver, whereas Φ_{lap} , Φ_{smooth} , and Φ_{clock} retain their roles as feasibility / regularization terms.

Practical Use. Stage 1 is used as the default recovery due to its stability and interpretability. Stage 2 is treated as an optional diagnostic refinement and is adopted only when it decreases the restoration-aware objective while maintaining admissibility. In practice this means checking

$$J_{\text{rec}}(\Gamma^{(2)}) < J_{\text{rec}}(\Gamma^{(1)})$$

together with the required admissibility constraints, e.g.

$$\Phi_{\text{lap}}(\Gamma^{(2)}) \leq \epsilon_1, \quad \Phi_{\text{clock}}(\Gamma^{(2)}) \leq \epsilon_4.$$

Default Operating Point. For the paper-facing default configuration, the discrete reference bank is taken to have width

$$\Delta_{\text{bank}} = 20^\circ$$

around the central reference phase, and the restoration weight is set to

$$\beta = 16.$$

This operating point is chosen as the smallest anchor strength that already gives a stable Stage 1 surrogate in the validation sweep. Narrower banks such as 5° are treated as unstable narrow-bank regimes, while very large anchor weights such as $\beta = 128$ are treated as high-anchor regimes that overly suppress useful Stage 2 diagnostic variation. Wider operating points or stronger anchors are therefore relegated to sensitivity / ablation tables rather than used as the main paper default.

10.10 Hybrid Policy Layer for Encoded Baselines

For the encoded-QEC baseline experiments we compare two recovery candidates:

- $A :=$ syndrome-first decoded recovery,
- $B :=$ relational recovery from the admissible restoration pipeline.

On top of these two candidates we define three hybrid policy layers.

- **C_1 (rule-based conservative hybrid).** Start from the syndrome baseline A , but allow a switch to B through a rule-based objective tie-break or through explicit veto conditions when A appears structurally inconsistent.
- **C_2 (balanced hybrid).** Compare A and B using a balanced score that combines syndrome support, restoration-aware objective utility, and inadmissibility penalty. This policy can preserve A even when B is relationally attractive, provided that the syndrome prior still strongly favors A .
- **C_3 (safety-first hybrid).** Enforce the strongest safety filter and block switching unless inadmissibility evidence is sufficiently strong. This policy is intended to probe whether a separate hard-safety operating zone exists.

Accordingly, C_1 should be interpreted as the default rule-based hybrid, C_2 as a balanced anti-over-switch policy, and C_3 as a strict safety-first policy.

10.11 Empirical Operating Zones of C_1 , C_2 , and C_3

The full encoded regime map is aggregated over 240 regime cells after averaging over seeds. On this grid the preferred-policy counts are

$$\#(C_1) = 205, \quad \#(C_2) = 35, \quad \#(C_3) = 0.$$

Thus C_1 remains the dominant default policy, C_2 occupies a nontrivial local operating zone, and C_3 does not form an independent safety-first region on the present grid.

The C_2 -preferred cells form a strongly structured cluster rather than a scattered set of exceptions. All 35 preferred cells occur under the phaseflip code, and $25/35 = 71.4\%$ arise under dephasing or phaseflip noise. This immediately shows that the gain of C_2 is regime-specific and concentrated in phase-sensitive stochastic sectors rather than spread uniformly across all perturbation types.

The advantage of C_2 is not driven by more aggressive relational selection. On the contrary, the dominant empirical signature is the suppression of harmful over-switching. In the C_2 -preferred cells the mean logical-fidelity gains satisfy

$$\overline{\Delta F}_{C_1} \approx 0.0147, \quad \overline{\Delta F}_{C_2} \approx 0.0279,$$

while the mean rates of selecting the relational candidate are

$$\overline{\chi_B(C_1)} \approx 0.9314, \quad \overline{\chi_B(C_2)} \approx 0.4162.$$

Hence the improvement of C_2 comes from switching *less*, not from switching more: C_1 tends to over-select B in precisely those cells where a balanced policy should continue to preserve the syndrome baseline.

Decision-reason analysis supports the same interpretation. In the C_2 -preferred cells, C_1 is typically triggered by rule-based objective tie-breaks or veto-style overrides, whereas C_2 more often preserves A through balanced score preference or inadmissibility penalty. This means that the distinction between C_1 and C_2 is not merely numerical tuning; it reflects two genuinely different policy logics. C_1 resolves ambiguity by allowing objective-driven switching, while C_2 explicitly regularizes against unnecessary relational intervention.

Score decomposition further shows that the balanced hybrid remains tilted toward preserving the syndrome baseline in these regimes. In the C_2 -preferred cells the mean syndrome support for A is high,

$$\overline{\text{Syn}(A)} \approx 0.9599,$$

the mean score difference favors retention of A ,

$$\overline{\Delta S} \approx -0.1275,$$

and the restoration-aware utilities of A and B remain closely matched. Thus the gain of C_2 does not arise because B is overwhelmingly better, but because the balanced score prevents a fragile objective tie-break from overpowering strong syndrome evidence.

The location of the C_2 -preferred cells is also informative. They occur primarily in low-to-mid strength / depth regions of the phase-sensitive stochastic regime, i.e. near the boundary where C_1 begins to overreact but the evidence for abandoning the syndrome baseline is still mixed. Accordingly, C_2 should be interpreted as a *local policy correction*, not as a global replacement for C_1 .

Interpretation. Taken together, these results indicate that C_2 occupies a meaningful local operating zone in phase-sensitive stochastic regimes by preventing harmful relational over-switching that is not adequately controlled by C_1 . The present grid therefore supports the following taxonomy:

- C_1 : default policy on most regimes,
- C_2 : balanced phase-sensitive correction,
- C_3 : no independent operating zone yet observed.

11 Clock Geometry and Relational Consistency

11.1 Clock Orbit and Local Distinguishability

Let the clock states be generated by

$$|\tau\rangle_C = e^{-i\hat{H}_C\tau}|0\rangle_C, \tag{93}$$

where $\hat{H}_C$ is the clock Hamiltonian.

We define the local distinguishability between adjacent clock labels as

$$d_{\text{clock}}^{\text{geom}}(\tau, \tau + \delta\tau) := 1 - |\langle \tau | \tau + \delta\tau \rangle_C|^2. \quad (94)$$

Equivalently, using the Bures distance,

$$D_B(\tau, \tau + \delta\tau) = \sqrt{2 \left(1 - \sqrt{F_\tau}\right)}, \quad (95)$$

where

$$F_\tau = F(|\tau\rangle\langle\tau|_C, |\tau + \delta\tau\rangle\langle\tau + \delta\tau|_C) = |\langle \tau | \tau + \delta\tau \rangle_C|^2. \quad (96)$$

For sufficiently small $\delta\tau$, we obtain the expansion

$$d_{\text{clock}}^{\text{geom}}(\tau, \tau + \delta\tau) = (\Delta H_C)_\tau^2 (\delta\tau)^2 + O((\delta\tau)^4), \quad (97)$$

where

$$(\Delta H_C)_\tau^2 = \langle \tau | \hat{H}_C^2 | \tau \rangle - \langle \tau | \hat{H}_C | \tau \rangle^2. \quad (98)$$

Interpretation The quantity $d_{\text{clock}}^{\text{geom}}$ measures the intrinsic distinguishability of adjacent clock labels and induces a local geometric structure on the clock orbit. It does not depend on external time and serves as an intrinsic resolution measure of the relational clock.

11.2 Relational Response Between Clock Slices

Let $\rho_S(\tau)$ be the clock-conditioned system state. We define the relational response between adjacent slices as

$$d_{\text{resp}}(\tau, \tau + \delta\tau) := d(\rho_S(\tau), \rho_S(\tau + \delta\tau)), \quad (99)$$

In the main formulation, we instantiate d_{resp} as the spectral response

$$d_{\text{resp}}(\tau, \tau + \delta\tau) := \|\Lambda(\tau + \delta\tau) - \Lambda(\tau)\|_2,$$

thereby aligning the clock-consistency penalty with the geometry of the spectral trajectory space. The fidelity-based alternative is discussed in the appendix as a robustness check.

11.3 Clock Consistency Measure

We define the clock consistency ratio as

$$\mathcal{C}_{\text{clock}}(\tau) = \frac{d_{\text{resp}}(\tau, \tau + \delta\tau)}{d_{\text{clock}}^{\text{geom}}(\tau, \tau + \delta\tau) + \epsilon}, \quad (100)$$

where $\epsilon > 0$ is a small regularization constant.

Interpretation This ratio measures whether the variation of the system state is consistent with the intrinsic resolution of the clock.

- $\mathcal{C}_{\text{clock}}(\tau) \ll 1$: stable relational encoding
- $\mathcal{C}_{\text{clock}}(\tau) \approx 1$: consistent evolution
- $\mathcal{C}_{\text{clock}}(\tau) \gg 1$: anomalous or inconsistent slicing

11.4 Trajectory-Level Clock Consistency

We extend the definition to the full trajectory:

$$\Phi_{\text{clock}}(\Gamma) = \int_{\mathcal{T}} \mathcal{C}_{\text{clock}}(\tau) d\mu(\tau), \quad (101)$$

or in discrete form,

$$\Phi_{\text{clock}}(\Gamma) = \sum_{r=1}^{m-1} \frac{d(\rho_S(\tau_r), \rho_S(\tau_{r+1}))}{d_{\text{clock}}^{\text{geom}}(\tau_r, \tau_{r+1}) + \epsilon}. \quad (102)$$

A trajectory is considered clock-consistent if

$$\Phi_{\text{clock}}(\Gamma) \leq \eta. \quad (103)$$

Remark 11.1. The admissibility threshold η is calibrated statistically from the background distribution of Φ_{clock} evaluated on unperturbed reference trajectories, whereas the slice spacing $\delta\tau$ is constrained by the clock resolution bound of Lemma 4.1. These two quantities serve distinct roles and are not directly related.

The abstract clock penalty Φ_{clock} introduced in Eq. (89) is concretely instantiated in Sec. 11 through the clock-geometry-based consistency ratio in Eqs. (99)–(101).